



Circulating MicroRNA ratios as diagnostic biomarkers for long COVID with an independent validation cohort



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Abstract

Background Long COVID is a multisystemic condition with heterogeneous clinical presentations lacking diagnostic biomarkers with discriminatory capacity. This study screened circulating microRNAs as candidate biomarkers for long COVID detection by miRNA-sequencing, followed by targeted qPCR quantification focusing on miRNA ratios to reduce technical variability.

Methods Overall, 206 long COVID patients (> 12 weeks after acute infection) and 71 uninfected, unvaccinated healthy controls were recruited through our prospective POSTCOV registry. Candidate microRNAs were screened by microRNA sequencing and quantified by targeted quantitative polymerase chain reaction, focusing on microRNA ratios expressed as cycle-threshold differences. In a training cohort, we developed a logistic regression model based on two microRNA ratios with age and sex as covariates. Internal validation was evaluated using cross validation and bootstrap. We then test the model in an independent external validation cohort processed with the same normalization and ratio definition.

Results Here we show both microRNA ratios (hsa-miR-484/hsa-miR-126-3p, hsa-miR-345-5p/hsa-miR-223-3p) are inversely associated with long COVID. Discrimination in the training cohort is moderate (AUC: 0.760), with acceptable calibration (intercept ~0, slope ~1, Brier score 0.131). In external validation, discrimination decreases (AUC: 0.676) and calibration reveals systematic risk overestimation (intercept -1.88, slope 0.77). Nevertheless, both ratios are consistently lower in long COVID compared to controls.

Conclusion Consistent between-group ratio differences support these microRNAs as candidate biomarkers, whereas limited external model performance highlights the challenges of transportability in heterogeneous long COVID cohorts. Target enrichment indicates inflammatory signaling pathways and integration with drug perturbation profiles suggests compounds for future validation and repurposing studies.

Plain Language Summary

Long COVID is a serious condition resulting from SARS-CoV-2 infection. It is associated with persistent symptoms affecting multiple organ systems, yet objective blood-based markers that could support clinical assessment remain limited. This study evaluates whether small molecules measurable in blood, known as microRNAs, differ between individuals with long COVID and non-long COVID controls. We used a stepwise approach: an initial screen to identify candidate microRNAs, confirmation by a method called RT-qPCR, and testing of a microRNA ratio-based prediction model in an independent validation cohort. Several microRNA ratios show consistent differences between long COVID and controls, while the model performance was decreased in external validation, suggesting sensitivity to cohort and/or technical differences. These findings support circulating microRNAs as a promising signal for future diagnostic support, but further validation in larger, well-matched populations is required before clinical implementation.

In May 2023, the World Health Organization (WHO) stated that after three years of the COVID-19 pandemic, the disease had become an established, though ongoing, public health problem, but did not represent a global health crisis anymore¹. The number of newly diagnosed COVID-19 cases has been gradually decreasing worldwide, but in the aftermath, a group of people is

left, who suffer from long lasting symptoms affecting multiple organ systems after a SARS-CoV-2 infection². This post-acute sequelae SARS-CoV-2 infection (PASC), also known as “long COVID” (LC), is a new syndrome, and although shares some similarities with other known post-viral syndromes, still requires further investigations to elucidate pathomechanisms,

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disease course, therapeutic options, and prognostic outlook³. The general consensus regarding the definition of long COVID is the persistent burden of symptoms at 12 weeks after a SARS-CoV-2 infection, irrespective of whether individuals failed to recover from their initial symptoms or experienced newly emerging symptoms following an apparent convalescent period. Diagnosis of long COVID is mainly achieved by the process of elimination of other illnesses that may cause similar symptoms⁴. However, due to the heterogeneous presentation forms and different clusters of symptoms, this approach requires generous time and resources. Therefore, the lack of diagnostic biomarkers and therapeutic targets for long COVID is an ongoing research gap requiring further studies⁵. For some LC presentation types (mainly neurological or pulmonary symptoms), interleukin-6 (IL-6), C-reactive protein (CRP), and tumor necrosis factor alpha (TNF- α) were proposed as such blood biomarkers⁶.

MicroRNAs (miRNAs), short single-stranded non-coding RNA molecules usually around 17 to 22 nucleotides long, are stably expressed in the blood and could be used as liquid biopsies⁷. This makes them interesting as potential diagnostic biomarkers due to their involvement in many regulatory and structural housekeeping pathways⁸. Moreover, miRNA ratios from both tissue and circulating miRNAs have been developed as diagnostic tools for discriminating melanoma from nevi or breast cancer screening^{9,10}. In chronic heart failure, selected miRNAs have been extensively studied as biomarkers to predict future hospitalisation¹¹. In acute COVID-19 infection, some miRNAs have been found to be associated with disease severity and mortality in hospitalized COVID-19 patients¹². Other studies have also implicated potential miRNAs for their role in the pathomechanisms of pain symptoms in long COVID¹³.

Beyond the potential as a biomarker, miRNAs may also offer mechanistic insights into disease processes and epigenetic regulation, which can be useful for informing therapeutic strategies by revealing candidate compounds that modulate relevant pathways. Lately, advances in transcriptomics and machine learning have enabled drug repurposing efforts by matching disease-related gene expression signatures with known pharmacological profiles, thus accelerating the identification of therapeutic leads from existing compounds¹⁴. As of yet, no specific drug has been approved for the treatment of long COVID, due to the lack of knowledge on the exact disease mechanisms. Given the hypothesized role of persistent endothelial dysfunction, coagulation abnormalities and prolonged inflammation in long COVID, antiviral, immune modulation, anticoagulation, antiplatelet and inhibition of neutrophil extracellular trap (NET) formation have been proposed as possible therapies¹⁵. Hence, the need for evidence-based therapeutic options besides supportive care and cardiopulmonary rehabilitation remains pertinent, particularly for working-age adults whose quality of life is gravely affected¹⁶; in this context, the integration of transcriptomics-informed drug repurposing into evolving treatment frameworks offers a promising outlook to lessen long COVID symptom burden.

This study aims to identify differentially expressed circulating miRNAs in long COVID patients with predominant cardiovascular, pulmonary, or neurological symptoms using a two-step approach consisting of miRNA-seq followed by RT-qPCR and additional external validation of the constructed miRNA ratio prediction model with a test cohort. Across cohorts, the key miRNA ratios remain directionally consistent between long COVID and controls, supporting their potential as reproducible biomarker signals. Predicted mRNA targets of the prioritized miRNAs implicate pathways related to immune signaling, vascular biology and cellular stress responses, which provides mechanistic hypotheses for long COVID pathophysiology.

Methods

Study population

This study was conducted in compliance with the Helsinki Declaration and with approval of the local ethics review board of the Medical University of Vienna (EC registration number: 1008/2021 approval date: 24 February 2021, and EC registration number: 1758/2022 approval date: 11 November 2022, ClinicalTrials.gov Identifier: NCT05398952). Study participants were recruited consecutively and gave their written informed consent. Long

COVID patients were prospectively enrolled in the Vienna PostCOV Registry between March 2021 and April 2023 from our specialized long COVID outpatient clinic, while the healthy controls were recruited between May 2020 and June 2020, who were all COVID-naïve and non-vaccinated (EC registration number: 1387/2020 approval date: 20 April 2020).

Adult patients with long COVID symptoms after a definitive infection with SARS-CoV-2 (confirmed by PCR test), after a minimum of 12 weeks of acute infection, were included in this study and further characterized into three subgroups according to their main symptom burden: cardiovascular (Cardio), pulmonary (Pulmo) and neurological/neuropsychiatric (Neuro) (Study flowchart: Fig. 1). These symptom groups were formed based on reports from existing literature and in accordance with the core outcome set for post-COVID-19 condition of the international Delphi consensus study^{17,18}. Long COVID patients with chest pain, palpitations, abnormalities in left ventricular or right ventricular pump function, chronic pericarditis/myocarditis proven by cardiac magnetic resonance imaging, or new heart rhythm disorders without respiratory abnormalities (normal chest X-ray, lung function and chest computer tomography) were classified as the Cardio group. Pulmo group patients were described as having resting or exertional dyspnea with pathological lung function or residual opacity in chest CT, without cardiac manifestation of long COVID syndrome. Patients with no objective organ disease pertaining to heart or lung were characterized into the Neuro group, if their main symptoms involved fatigue, headaches, concentration difficulties, vegetative symptoms and paresthesia. Exclusion criteria were non-willingness to participate or withdrawal of informed consent, participation in any SARS-CoV-2 pharmacological trial, pregnancy, renal insufficiency (glomerular filtration rate <45 ml/min/1.73 m²) or systemic inflammatory disease.

Clinical parameters

As part of the standardized workup of long COVID patients at our outpatient clinic specialized for post-COVID conditions at the Department of Internal Medicine II, Division of Cardiology, of the Medical University of Vienna, type and severity of long COVID symptoms, date of COVID-19 infection(s), medical history and comorbidities, vital parameters, current medication and COVID vaccinations were recorded. Clinical findings and reports of blood and imaging tests performed prior to their baseline visit (such as lab work, spirometry, chest X-ray, cardiac MRI) were recorded, as well as the results of a venous blood draw and electrocardiogram (ECG) on the day of the baseline visit. Quality of life was assessed with the EQ-5D-3L questionnaire in German, including a visual analogue scale (VAS) for general quality of life (range 0 /worst/–100 /best/)¹⁹.

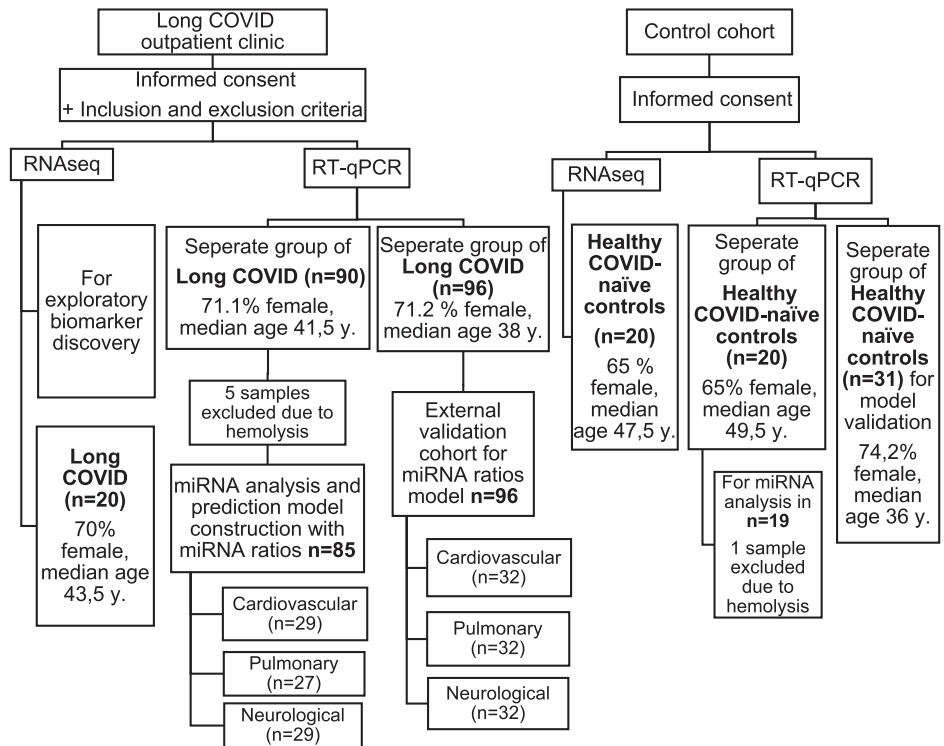
Sample collection and processing

Serum and EDTA-plasma samples were obtained through peripheral venipuncture at baseline and follow-up visits. EDTA-plasma samples were processed immediately by centrifugation and frozen at –80 °C until further analysis at the central biobank of the Medical University of Vienna. Serum samples were left standing to allow for clotting for 15 min before centrifugation for 15 min at 1500 g at 4 °C and then frozen at –80 °C until further analysis at the central biobank of the Medical University of Vienna. Blood count, coagulation, electrolytes, kidney function, liver function, cardiac enzymes and inflammatory parameters were processed and measured by standard routine protocols at the diagnostic laboratory of the Department of Laboratory Medicine, Medical University of Vienna. Standard routine diagnostic tests were conducted on a cobas e801 analyzer (Roche) at the diagnostic laboratory of the Department of Laboratory Medicine, Medical University of Vienna, to measure total protein, albumin, ferritin, CRP, high-density lipoproteins (HDL) and low-density lipoproteins (LDL).

RNA extraction

Total RNA was extracted from samples using Maxwell RSC miRNA Tissue Kit (Promega, Germany, cat. no. AS1460) in combination with the Maxwell RSC48 instrument. Samples were centrifuged for 5 minutes at 12000 g at 4 °C to remove cellular debris. For each sample, 200 μ L sample or 100 μ L

Fig. 1 | Study flowchart. Overall miRNAs were measured and compared between 201 long COVID patients and 70 healthy controls.



sample with 100 μ L water were homogenized with 200 μ L homogenization mix which consist of homogenization buffer and thioglycerol. Further 200 μ L lysis buffer and 15 μ L proteinase K were added. A synthetic RNA oligonucleotide mix obtained from the miRCURY Spike-In kit (Qiagen, Germany, cat. no. 339390) is added to each sample at equimolar amounts prior to RNA extraction. These spike-ins will be subsequently used to monitor RNA extraction efficiency. Samples were vortexed and incubated for 2 h at +37 $^{\circ}$ C followed by automated RNA extraction on the Maxwell RSC instrument. Finally, total RNA was eluted in 50 μ L nuclease free water and stored at -80° C until further analysis.

RNA sequencing and library preparation

Sequencing was performed in SR100 mode on Illumina's NovaSeq 6000 device using the SP flowcell. Overall quality of the next-generation sequencing (NGS) data was evaluated automatically and manually with fastQC v0.11.9 and multiQC v1.10^{20,21}. Reads from all passing samples were adapter trimmed and quality filtered using cutadapt v3.3 and filtered for a minimum length of 17nt²². Mapping steps were performed with bowtie v1.3.0²³ and miRDeep2 v2.0.1.2²⁴, whereas reads were mapped first against the genomic reference GRCh38.p12 provided by Ensembl²⁵ allowing for two mismatches, and subsequently²⁶, filtered for miRNAs of hsa (*Homo sapiens*) only, allowing for one mismatch. Statistical analysis of pre-processed NGS data was done with R v4.0²⁷.

Analysis of differentially expressed microRNAs

At study design, no data were available regarding miRNAs associated to long COVID disease, so the sample size and power calculation could only be estimated based on existing miRNA literature data in other diseases²⁸. The discovery sequencing phase was therefore planned as an exploratory screen informed by prior small RNA-seq studies and intended to nominate candidates for downstream verification. With 20 participants per group, a comparison at two-sided $\alpha = 0.05$ provides approximately 80% power to detect a standardized mean difference of Cohen's $d \approx 0.91$, which corresponds to an expression difference of about ~ 1.5 -fold change. Differential expression in the discovery phase was assessed using false discovery rate (FDR) control at 0.2 for a balance of type I and type II error rates. Candidate

miRNAs taken forward to RT-qPCR were selected based on a combination of (i) FDR-significant differential expression and effect size ranking in the discovery phase and (ii) a priori biological or clinical relevance supported by published evidence, even if not meeting the FDR threshold in discovery. The validation phase was therefore considered confirmatory for the selected candidates. Differential expression (DE) analysis with edgeR v3.32 used the quasi-likelihood negative binomial generalized log-linear model functions provided by the package²⁹. The independent filtering method of DESeq2 was adapted for use with edgeR to remove low abundant miRNAs and thus optimize the FDR correction³⁰. A logarithmic fold change (logFC) of $\geq |0.5|$ and a FDR of < 0.05 were considered statistically significant for sequenced miRNAs for subsequent mRNA target prediction. Heatmaps were generated with *ph heatmap* (version 1.0.12).

Target prediction

MiRTarBase (Release 9.0)³¹ and Target Scan (http://www.targetscan.org/vert_80/)³² were used to predict the mRNA targets of the differentially expressed miRNAs. Targets were included for further analysis if they were validated by a strong evidence method in miRTarBase or if they had a cumulative weighted context ++ score of ≤ -0.5 in Target Scan.

For visualization, Cytoscape (version 3.9.1)³³ was used to graph the network.

Gene set enrichment analysis

We used the web server Enrichr (<https://maayanlab.cloud/Enrichr/>, last accessed on 28 March 2025) to perform pathway enrichment analysis by inputting the genes from target prediction, consolidating the Reactome, MSigDB Hallmark and WikiPathways databases. The top ten pathways with an adjusted p -value < 0.05 and counts > 2 were visualised^{34–36}. The protein-protein interaction (PPI) network analysis was performed using STRING (<https://string-db.org/>, last accessed on 03 Jan 2025)³⁷.

Reverse transcription (RT) and quantitative polymerase chain reaction (qPCR)

Starting from total RNA samples (as described above in 2.4), cDNA was synthesized using the miRCURY RT Kit (Qiagen, Germany, cat. no.

339340). Reaction conditions were set according to recommendations by the manufacturer, and 2 μ L of total RNA was used as input in a 10 μ L reaction. Quantitative PCR reactions were set up using miRCURY SYBR[®] green master mix (Qiagen, cat. no. 339345) with a 1:100 diluted cDNA and commercial LNA-enhanced primer assay (Qiagen, cat. no. 339306). Reactions were performed in a 384-well primer spotted plates in a Roche LC480 II instrument (Roche, Germany) with the following temperature settings: 95 °C for 2 minutes, 45 cycles of 95 °C for 10 s and 56 °C for 60 seconds, followed by melting curve analysis. The second derivative maximum method was used to calculate the cycle of quantification values (Cq-values) and were normalized by the global mean across the samples included in the NGS analysis. Additionally, cel-miR-39-3p, Unisp2 and Unisp3 were added as cDNA, RNA and PCR spike-ins. Hemolysis was assessed using the ratios of miRNA-23a-3p and miRNA-451a, and samples with hemolysis index values of above 7 were excluded from further analysis to reduce the potential risk of hemolysis-induced variation of miRNA levels³⁸. Due to high hemolysis index, 5 samples from the long COVID cohort and 1 sample from the healthy controls were excluded for further analysis.

MiRNA ratios, random forest analysis and logistic regression model

MiRNA ratios (Δ Ct) were formed for small RNA-seq and RT-qPCR samples and compared with the Mann-Whitney U test with adjustment for multiple testing using Benjamini-Hochberg FDR between long COVID and healthy groups. The miRNA ratios dataset was initially assessed for missing values, with columns containing more than 10% missing data removed. The remaining missing values were imputed using median imputation, due to skewed distributions of the miRNA ratios. Weighted random forest analyses were performed (R package *randomForest*, version 4.7-1.2) with standard settings and 1000 trees to identify the most important miRNA ratios in the classification of patients with long COVID compared to those without long COVID³⁹. Feature importance was assessed using the mean decrease in Gini impurity. The model's predictive performance was evaluated using the area under the receiver operating characteristic (ROC) curve (AUC).

Logistic regression models were assessed using miRNA ratios as predictors. The final logistic regression model included the top two important miRNA ratios (hsa-miR-484/hsa-miR-126-3p and hsa-miR-345-5p/ hsa-miR-223-3p) along with age and sex as covariates. Odds ratios (OR) were reported per 1 Ct increase with 95% confidence intervals. The Variance Inflation Factor (VIF) was <3 for all variables in the regression models, indicating minimal multicollinearity. Model fit was summarized using the Akaike Information Criterion (AIC), and nested model comparisons were performed using likelihood-ratio tests (R package *caret*, version 6.0-94). Discrimination was assessed using ROC curve analysis and AUC (R package *pROC*, version 1.18.5). For descriptive classification, a confusion matrix was generated to assess sensitivity, specificity, likelihood ratios, positive predictive value, negative predictive value and accuracy at the threshold of 0.80. Internal validation of the final model was performed using 5-fold cross-validation and nonparametric bootstrap resampling (R = 1000) (R package *boot*, version 1.3-28.1). Calibration was assessed using calibration-in-the-large (CITL/intercept), calibration slope, and Brier score. Calibration was additionally visualized by grouping predictions into deciles of predicted risk and comparing mean predicted probabilities to observed event rates.

A separate group of 96 long COVID patients and 31 healthy controls were included as an independent validation cohort (test group). External validation was performed by applying the final logistic regression model, constructed on the RT-qPCR miRNA values, to the test group without refitting. Discrimination was quantified by AUC with bootstrap 95% confidence intervals. Calibration was assessed using CITL, calibration slope, and Brier score, supported by a calibration plot (R package *rms*, version 6.7-1). Logistic recalibration by updating the intercept and slope of the original linear predictor was calculated in the validation cohort as a sensitivity analysis.

Identification of therapeutic targets by drug signature enrichment analysis

To identify candidate compounds capable of reversing the gene expression signature regulated by differentially expressed miRNAs, we used the LINCS L1000 Chemical Perturbation Consensus Signatures (Chem Pert) dataset via the Enrichr platform (<https://maayanlab.cloud/Enrichr/>)^{40,41}. The input gene set comprised upregulated target genes inferred from differentially expressed miRNAs from the validation cohort. Compounds were ranked based on their adjusted *p*-values and combined enrichment scores.

Statistical analysis

Statistical analyses were performed with R (version 2022.12.0 + 353). Data are presented as mean $\pm$ standard deviation (normal distribution) or median (interquartile range, IQR). The normal distribution assumption was tested with the Shapiro-Wilks test. One-way ANOVA of means was used for post-hoc analysis comparing differences between groups. The Benjamini-Hochberg method was applied to correct for multiple testing (= false discovery rate, FDR). A two-tailed *p*-value < 0.05 was considered statistically significant.

Results

Patient characteristics

Overall, the median age of the 110 long COVID patients (71% female) and 40 healthy controls (65% female) was 40.5 and 50 years, respectively. The patient cohort consisted mainly of non-smokers (91.5%), and most had a mild course of COVID-19, not requiring hospitalisation (92.7%). The median time after COVID-19 infection was 244 days. Seventy-four patients (67.3%) had received a COVID vaccination before the blood draw at our specialized long COVID clinic. Long COVID patients reported a moderate decrease in general quality of life with a median VAS score of 40 out of 100 (quality of life scale: 0 = worst, 100 = best), while healthy controls stated a median of 85.5 out of 100.

For the first step of biomarker discovery by small RNA-seq, we included samples of 20 LC patients and 20 controls, while as a second step the other 90 LC samples and 20 controls were used for validation. Supplementary Table S1 and Supplementary Table S2 provide detailed descriptive characteristics of the long COVID patients (*n* = 20) and healthy controls (*n* = 20) for discovery and a separate independent validation cohort (90 LC and 19 healthy controls). The general distribution of clinical parameters was similar between patients of the small RNA-seq and the RT-qPCR cohort.

Discovery cohort via miRNA-sequencing

A total of 1859 miRNAs were identified and mapped from small RNA-sequencing, while the mean number of distinct identified miRNAs was 745 across all sequenced samples. The heatmap shows the top differentially expressed miRNAs with FDR < 0.2 clustered using correlation distance and average linkage with applied unit variance scaling (Fig. 2A). The first two principal components graphed depict the samples grouped by their distinct miRNA expression (Fig. 2B). Small RNA-seq revealed five deregulated miRNAs (hsa-miR-484, hsa-miR-21-3p, hsa-miR-345-5p, hsa-miR-1307-3p, hsa-miR-203a-3p) in long COVID versus Healthy groups. Detailed data on log₂FC and FDR of the DE-miRNAs can be found in Suppl. Table S3. For upregulated miRNA in long COVID versus Healthy, 114 mRNA targets were detected (Fig. 2C, Suppl. Table S4). The STRING analysis of upregulated miRNAs in long COVID patients showed IL-10 as the central node in protein-protein interaction (PPI) network with TGFB1, SMAD7, ACVR1B, STAT3, CDKN1A, CCL3 and further interactions between PRKCZ and PTEN, USE1 and RINT1, MAT2B and MAT2A (PPI enrichment *p*-value = 0.0202) (Fig. 2D). Pathway enrichment analysis of these predicted mRNA targets revealed "Apoptosis", "Signaling by ALK fusions and activated point mutants", "spinal cord injury", "IL-6/JAK/STAT3 Signaling", "SARS CoV2 innate immunity evasion and cell immune response", "AGE RAGE pathway", as enriched pathways for the provided gene sets. The top 10 enriched pathways are shown in Fig. 2E.

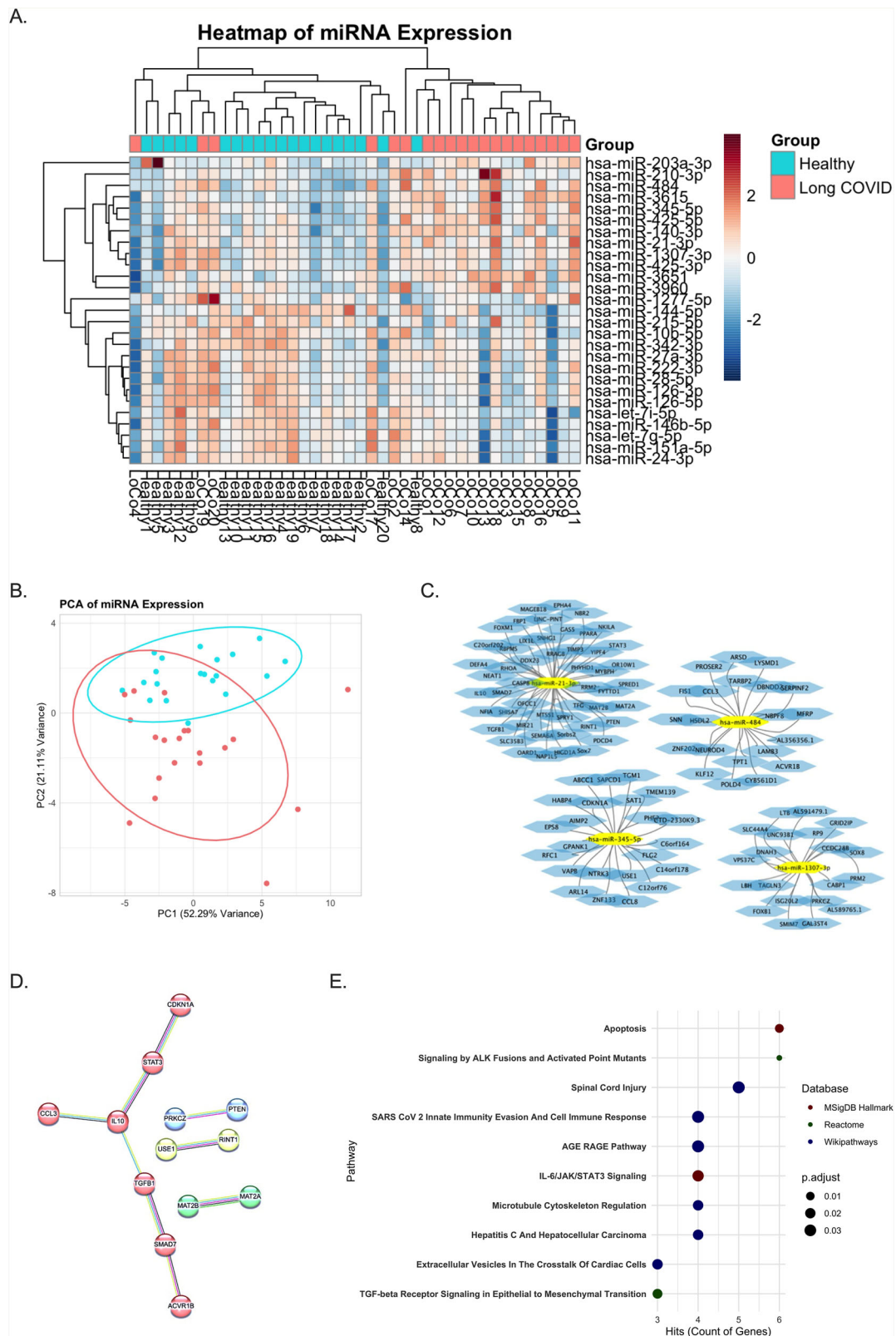


Fig. 2 | miRNA-seq results of 40 samples (20 long COVID, 20 healthy controls). **A** Heatmap of miRNA expression assessed by small RNA-seq. **B** Principal component analysis of miRNA expression assessed by small RNA-seq. Healthy controls = blue, long COVID patients = red. **C** 114 mRNA targets of upregulated miRNAs

were identified in Long COVID (= long COVID patients) compared to Healthy (= COVID naive controls). **D** Protein-protein interaction (PPI) network of mRNA of deregulated miRNAs. **E** Pathway enrichment analysis using multiple databases of predicted target genes.

Validation cohort via miRNA RT-qPCR

For a comprehensive study of miRNAs in long COVID patients, an additional 90 serum samples from an independent (not previously analyzed) long COVID cohort were subjected to measurement of miRNA expression with RT-qPCR, along with 20 healthy controls. MiRNAs were selected due to either their presence in miRNA-seq as differentially expressed or based on previously reported interesting aspects found in the literature (Suppl. Table S5). Due to the hemolysis index of above 7, five samples from the long COVID cohort and one control sample were excluded for further analysis.

Three miRNAs were significantly (adj. $p < 0.05$) downregulated in long COVID patients by RT-qPCR measurement: hsa-miR-210-3p, hsa-miR-345-5p, and hsa-miR-484 (Fig. 3A), while hsa-miR-9-5p was upregulated in the LC group compared to controls (Table 1). In subgroup analyses (Suppl. Figs. S1–S3), the overall pattern of lower miRNA abundance in long COVID was preserved across cardiovascular, neurologic, and pulmonary phenotypes. Compared with controls, the Cardio subgroup showed significantly lower levels of hsa-miR-210-3p ($p < 0.01$), hsa-miR-345-5p ($p = 0.04$) and hsa-miR-484 ($p = 0.03$), alongside a modest increase in hsa-miR-122-5p ($p = 0.02$) (Fig. S1). The Neuro subgroup likewise displayed reduced hsa-miR-210-3p ($p = 0.02$) and hsa-miR-484 ($p = 0.03$) relative to controls (Fig. S2). In the pulmonary subgroup, hsa-miR-345-5p ($p = 0.01$) and hsa-miR-484 ($p = 0.02$) remained lower than in controls, while hsa-miR-9-5p was higher ($p < 0.01$) (Figure S3).

MiRNA ratios for predicting long COVID and internal model validation

MiRNA ratios were constructed separately for RNA-seq and RT-qPCR cohorts. 208 miRNA ratios were significantly different between long COVID and healthy individuals in the RNA-seq cohort, while only two miRNA ratios were for the RT-qPCR group (Fig. 3B). The two miRNA ratios were miR-484/miR-126-3p and miR-126-3p/ miR-484. Using a random forest classifier the model classified long COVID cases with very low class error with a classification error of 1.18%, but performed poorly for controls with 78.95% of the healthy controls as false positives (Fig. 3C). The random forest prediction model revealed miR-484/miR-126-3p and miR-345-5p/miR-223-3p as the top two most important variables (Fig. 3D) and had an AUC of 0.714 (95% CI: 0.585–0.842, DeLong)(Fig. 3E). The out-of-bag estimate of error rate was 15.38%.

Logistic regression model development and internal validation were conducted in the training cohort ($n = 104$; LC prevalence 0.817). The AUC for predicting long COVID with hsa-miR-484/hsa-miR-126-3p, hsa-miR-345-5p/hsa-miR-223-3p and hsa-miR-210-3p/hsa-miR-126-3p by themselves was 0.77, 0.69 and 0.71, respectively (Fig. S4). Model 1 included the top two miRNA ratios hsa-miR-484/hsa-miR-126-3p and hsa-miR-345-5p/hsa-miR-223-3p from the random forest analysis (AIC: 94.14), while Model 2 used the top two miRNA ratios with age and sex as covariates (AIC: 94.86). Due to the small sample size and to avoid overfitting, cross-validation and bootstrap analyses were performed. In Model 1, both miRNA ratios showed inverse directions with LC (Table 2). Higher Ct values of hsa-miR-484/hsa-miR-126-3p were directionally inverse to the odds of long COVID (OR = 0.516, 95% CI 0.214–1.162), and likewise for hsa-miR-345-5p/hsa-miR-223-3p (OR = 0.600, 95% CI: 0.295–1.147). Discrimination was moderate with an AUC = 0.752, and sensitivity was 70.6% and specificity 68.4% (Fig. 4A).

In Model 2, after adjustment for demographic covariates, both miRNA predictors remained directionally inverse with LC: OR 0.487 (95% CI: 0.200–1.102) for hsa-miR-484/hsa-miR-126-3p and OR 0.619 (95% CI: 0.298–1.210) for hsa-miR-345-5p/hsa-miR-223-3p. Age showed OR 0.964 per year (95% CI 0.913–1.013), and sex showed no association (OR 0.987; 95% CI: 0.296–3.590). Adding age and sex did not significantly improve fit compared with the miRNA ratios-only model (nested likelihood-ratio test $p = 0.332$). However, the variables were kept as covariates for this final model. Discrimination for the final model in the training cohort was AUC = 0.760 (Fig. 4B). At a probability cutoff of 0.80, sensitivity was 72.9% (95% CI: 62.2–82.0) and specificity 57.9% (95% CI: 33.5–79.8). Table 3

depicts the confusion matrix including sensitivity, specificity, positive and negative likelihood ratios, positive and negative predictive values (PPV, NPV), and accuracy of the model at different disease prevalence percentages within the range of reported long COVID prevalence (10%, 6%, and 3%). At 10% prevalence, PPV was 16.1% (95% CI: 10.12–24.9) and NPV was 95.1% (95% CI: 92.0–97.0). With lower prevalence, PPV decreased to 10.0% (6%) and 5.1% (3%), while NPV increased to 97.1% and 98.6%, respectively (Table 3).

In 5-fold cross-validation, mean ROC AUC was 0.772 and mean PR AUC was 0.925. Bootstrap internal validation ($R = 1000$) yielded an apparent AUC of 0.760 (95% CI: 0.680–0.901) (Fig. 4C). Calibration in the training cohort was acceptable on average: CITL approximately 0 (95% CI –0.513 to 0.564), calibration slope approximately 1 (95% CI 0.428–1.704), and Brier score 0.131.

External validation and sensitivity analysis

External validation was performed in an independent cohort ($n = 127$; LC $n = 96$, Control $n = 31$), which had comparable characteristics to the initial RT-qPCR cohort. Detailed baseline characteristics for this additional cohort are depicted in Suppl. Table S6. Model coefficients and decision threshold were fixed on the training cohort prior to evaluation in the external validation cohort. On external validation, discrimination was moderate (AUC = 0.676; bootstrap 95% CI: 0.564 to 0.792) (Fig. 4D). Calibration assessment showed systematic overprediction (CITL –1.74, 95% CI: –2.16 to –1.28) and a calibration slope below 1 (0.774; 95% CI: 0.276–1.318). The Brier score was 0.212. Predicted probabilities were generally high (decile mean predicted risk approximately 0.83–0.99), and at a probability cutoff of 0.80, sensitivity was 97.9% and specificity 3.2% (TP 94, FN 2; TN 1, FP 30), indicating limited transportability of the absolute probability scale of the logistic regression model to separate cohorts. As an exploratory analysis, the ROC-derived best threshold in the validation cohort was assessed and would be 0.957 with a sensitivity of 63.5% and specificity of 74.2%.

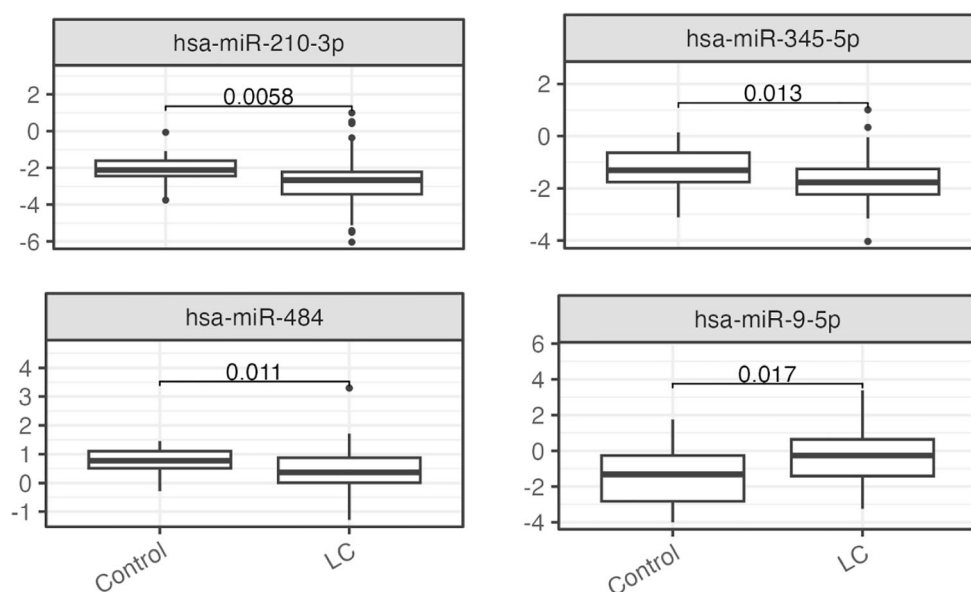
Logistic recalibration of the linear predictor yielded a negative recalibration intercept (–1.225) and a recalibration slope < 1 (0.774), consistent with overconfident predictions. Recalibration improved overall probability accuracy as reflected by a lower Brier score of 0.171. Calibration plots can be found in Suppl. Fig. S5.

Nevertheless, both miRNA ratios that define the model showed a consistent directional shift between long COVID and controls across cohorts. For hsa-miR-484/hsa-miR-126-3p, values were lower in long COVID than in controls in the training cohort (mean –3.231 vs. –2.651) and this separation persisted in the test cohort (mean –4.349 vs. –4.009) with comparable standard deviations (Fig. 3E). Likewise, for hsa-miR-345-5p/hsa-miR-223-3p, long COVID showed lower values than controls in training (mean –8.857 vs. –8.148) and in the test cohort (mean –9.811 vs. –9.097) (Fig. 3F). Median and interquartile ranges followed the same pattern for both ratios, indicating that the shift between the cohorts was not driven by a small number of extreme values.

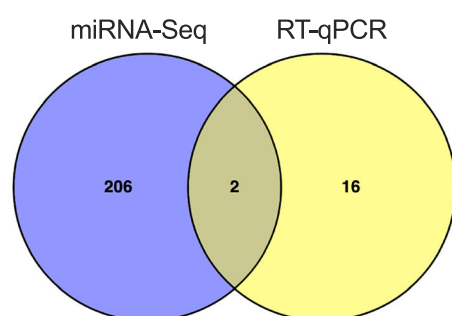
Drug target prediction based on miRNA signature

A list of target genes for these deregulated miRNAs used for drug target prediction can be found in Supplementary Table S7. We identified compounds that are predicted to reverse the mRNA signature affected by the downregulated miRNAs in long COVID patients through the LINCS L1000 Chemical Perturbation Consensus Signatures and grouped them by mechanistic class (Table 4). The full list of top 25 compounds from LINCS L1000 Chem Pert Consensus Sigs with their combined enrichment scores can be found in Suppl. Table S8. Several compounds of interest were found by connectivity analysis with potential to inverse the transcriptomic signature, which involved pathways relevant to chronic inflammatory and vascular stress. A miRNA-mRNA-drug network was constructed to visualize the interactions between molecules (Fig. 5). The affected pathways of mRNA target genes were significantly enriched in the p53 pathway and TNF-alpha signaling via NF-kB. Compounds acting as signal-transduction enzyme inhibitors or modulators contained 5-iodotubercidin (an adenosine

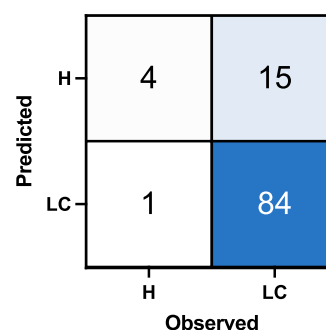
A.



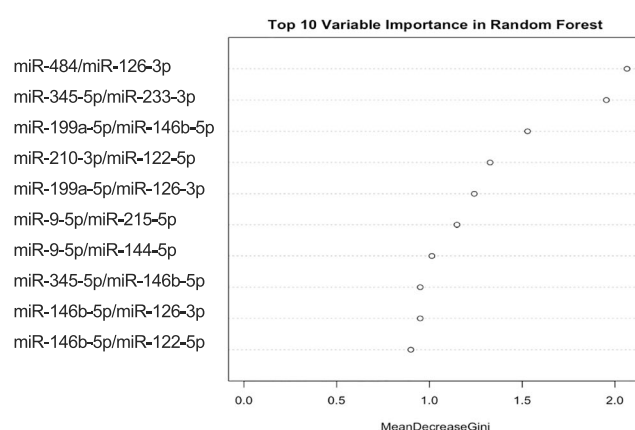
B.



C.



D.



E.

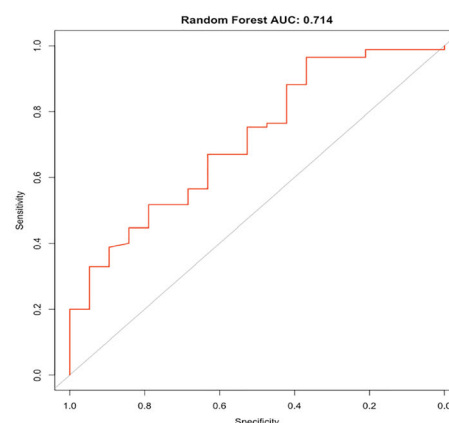


Fig. 3 | Assessment of miRNAs by RT-qPCR in the Training Cohort (104 samples: 85 long COVID, 19 healthy controls) and miRNA ratios for predicting long COVID. A Boxplots of significant differentially expressed miRNAs in the RT-qPCR training cohort, Ct values were inversed and global mean normalized. Median and interquartile range, outliers depicted as dots. Wilcoxon rank test, two-tailed *p*-value. **B** Venn diagram of miRNA ratios deregulated in miRNA-seq and RT-qPCR:

484/hsa-miR-126-3p and hsa-miR-126-3p/hsa-miR-484 were significantly different by both measurement methods. **C** Confusion matrix of random forest analysis classified long COVID (LC) patients compared to healthy controls (H). **D** Ten most important miRNA ratios from the random forest analysis plotted by decreasing MeanDecreaseGini value. **E** ROC of random forest model, AUC: 0.714.

Table 1 | Differences of miRNA Ct values between 85 long COVID patients and 19 healthy controls

miRNA	Median Ct value difference	P-value	adj.p (BH)	Median Ct value Control	Median Ct value LC	Control, N	LC, N
hsa-miR-210-3p	−0.55	0.01	0.05	−2.12	−2.67	19	85
hsa-miR-345-5p	−0.46	0.01	0.05	−1.31	−1.77	19	85
hsa-miR-484	−0.41	0.01	0.05	0.77	0.36	19	85
hsa-miR-9-5p	1.05	0.02	0.05	−1.32	−0.27	19	83
hsa-miR-23a-3p	0.13	0.06	0.16	3.82	3.95	19	84
hsa-miR-122-5p	1.02	0.07	0.17	−8.02	−7.00	19	84
hsa-miR-144-5p	−0.21	0.08	0.17	−1.12	−1.33	19	85
hsa-miR-22-5p	−0.36	0.17	0.30	−4.05	−4.41	18	81
hsa-miR-126-3p	−0.07	0.32	0.46	3.68	3.61	19	85
hsa-miR-215-5p	−0.27	0.30	0.46	−1.13	−1.39	19	85
hsa-miR-223-3p	0.17	0.34	0.46	6.93	7.10	19	85
hsa-miR-199a-5p	−0.02	0.46	0.57	−2.02	−2.04	19	85
hsa-miR-451a	−0.10	0.57	0.65	8.75	8.66	19	84
hsa-miR-146b-5p	−0.01	0.92	0.92	−2.49	−2.50	19	85
hsa-miR-203a-3p	0.41	0.92	0.92	−5.87	−5.47	6	46

BH Benjamini-Hochberg procedure, Ct cycle threshold, LC long COVID.
Ct values were inversed and normalised to the global mean. The miRNAs are ordered ascendingly according to the adjusted *p*-values.

Table 2 | Logistic regression models for long COVID vs. Healthy

Coefficients:	Odds ratio	CI lower (2.5%)	CI upper (97.5%)	P-value
Model 1:				
hsa-miR-484/hsa-miR-126-3p	0.516	0.214	1.162	0.121
hsa-miR-345-5p/hsa-miR-223-3p	0.600	0.295	1.147	0.136
Model 2:				
hsa-miR-484/hsa-miR-126-3p	0.487	0.120	1.102	0.085
hsa-miR-345-5p/hsa-miR-223-3p	0.619	0.298	1.210	0.164
Age	0.964	0.913	1.013	0.151
Male	0.987	0.296	3.590	0.984

Model 1: raw mode with only miRNA ratios, AIC: 94.15; Model 2: miRNA ratios + covariates age and sex, AIC: 95.94.

kinase inhibitor), ZG-10 (JNK inhibitor), cilostazol (phosphodiesterase-3 inhibitor), and L-744832 (farnesyl transferase inhibitor). Secaliferol (24 R,25-dihydroxyvitamin D3), a vitamin D metabolite, was also found related to the mRNA signature with a high connectivity score of 113.2. Further compounds were of the G-protein coupled receptor mechanistic class, consisting of clidinium (anticholinergic) and LY-288513 (a selective CCK2 receptor antagonist). Torcetrapib, a discontinued cholesteryl ester transfer protein inhibitor, was also found to share interactions with this miRNA-mRNA signature. Lastly, vidarabine was also represented in this network analysis, which is an antiviral nucleoside analog.

Discussion

In our case-control study from the Vienna POSTCOV registry, we identified several differentially expressed circulating miRNAs between long COVID patients and healthy controls. Target prediction and pathway enrichment analyses showed enriched genes involved in cell signaling, inflammation, angiogenesis, and T-cell regulation. In an independent RT-qPCR cohort, three miRNAs (hsa-miR-210-3p, hsa-miR-345-5p, hsa-miR-484) remained significantly lower in long COVID, while hsa-miR-9-5p was higher. Sub-group analyses suggested that the overall pattern of reduced miRNA abundance in long COVID was preserved across cardiovascular, neurologic and pulmonary phenotypes. Further, the miRNA ratio hsa-miR-484/hsa-miR-126-3p was identified as a discriminator between long COVID patients

and healthy controls in both the RNA-seq and the RT-qPCR cohort with consistently lower ratio values in long COVID patients compared to controls. We constructed a logistic regression model using hsa-miR-484/hsa-miR-126-3p and hsa-miR-345-5p/hsa-miR-223-3p for predicting long COVID and validated this model on an independent cohort, which showed moderate discrimination (AUC: 0.676), but some batch effect despite the same normalization procedures as to the training cohort.

Several of the miRNAs from our analysis, including hsa-miR-484, hsa-miR-345-5p, hsa-miR-210-3p, hsa-miR-144-5p, hsa-miR-199a-5p, hsa-miR-233-3p, and hsa-miR-126-3p, have been previously reported as regulators of immune function, inflammation, hypoxia, and vascular dysfunction^{42–46}. Notably, miR-484 has been associated with mitochondrial stress by targeting FIS1, a key mediator of mitochondrial fission. This interaction can influence cellular apoptosis and has been observed in various cell types, including cardiomyocytes and adrenocortical cancer cells⁴⁷. In the context of COVID-19, mitochondrial dysfunction and endothelial damage are prominent features, suggesting that dysregulation of miR-484 may contribute to these pathologies. We have previously published our findings of higher levels of autoantibodies against endothelin type A receptor (ETAR) in long COVID patients with orthostatic intolerance correlating to higher symptom burden, additionally supporting the hypothesis that endothelial dysfunction could play a role in the pathophysiology of long COVID⁴⁸.

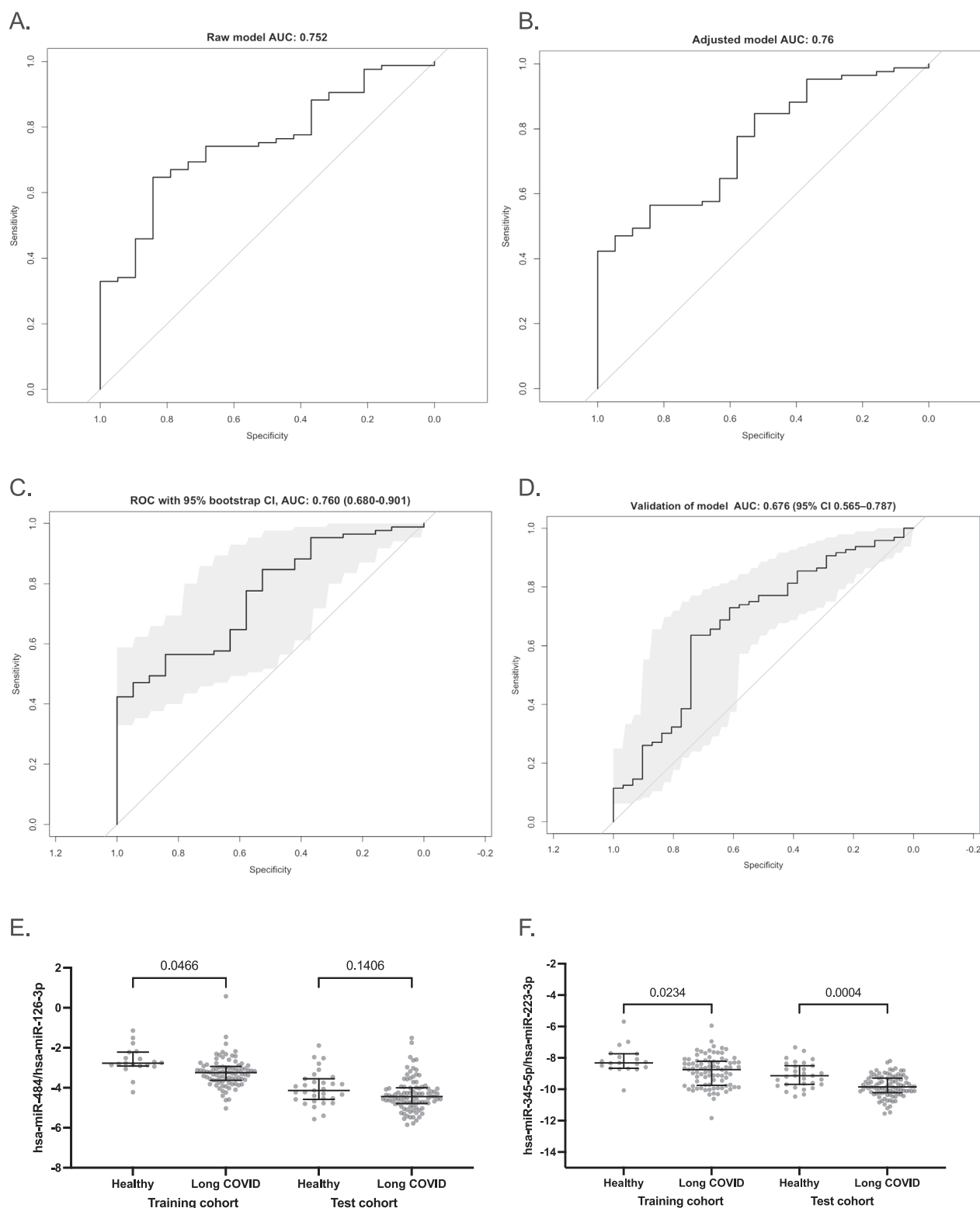


Fig. 4 | Logistic regression model with RT-qPCR miRNA ratios performance on the training cohort and test cohort for discrimination of long COVID. **A** Model 1 (hsa-miR-484/hsa-miR-126-3p, hsa-miR-345-5p/hsa-miR-233-3p): ROC curves of miRNA ratios to predict long COVID, AUC: 0.752. **B** Adjusted model (Model 2): Model 1 + covariates (age and sex), AUC: 0.760. **C** ROC with bootstrap resampling ($R = 1000$) of Model 2, AUC: 0.760 (95%CI: 0.680–0.901). **D** Model 2 test on external validation samples (96 long COVID, 31 controls), AUC: 0.676 (95%CI:

0.656–0.787). **E** Median miRNA ratio of hsa-miR-484/hsa-miR-126-3p is significantly lower in long COVID patients compared to healthy controls of the training cohort, but not in the test cohort, median and interquartile range, two-tailed p -value. **F** Median miRNA ratio of hsa-miR-345-5p/hsa-miR-233-3p is lower in long COVID patients compared to healthy controls in both training and test cohort, median and interquartile range, two-tailed p -value.

Table 3 | Confusion matrix for predicted classes of the logistic regression model (threshold 0.80)

	Healthy (observed)	Long COVID (observed)	Total (predicted)
Healthy (predicted)	11	23	34
Long COVID (predicted)	8	62	70
Total (observed)	19	85	104
Statistics	Value	95% CI	
Sensitivity	72.94%	62.21–82.01%	
Specificity	57.89%	33.50–79.75%	
Positive Likelihood Ratio	1.73	1.01–2.98	
Negative Likelihood Ratio	0.47	0.28–0.78	
Disease prevalence	10%	6%	3%
	Value (95%CI)	Value (95%CI)	Value (95%CI)
Positive Predictive Value (PPV)	16.14% (10.06–24.88)	9.96% (6.04–15.99)	5.09% (3.02–8.44)
Negative Predictive Value (NPV)	95.06% (91.98–97.00)	97.1% (95.23–98.25)	98.58% (97.63–99.15)
Accuracy	59.40% (49.33–68.92)	58.80% (48.72–68.36)	58.35% (48.27–67.94)

Table 4 | Summarized list of compound drug classes that are predicted to reverse the mRNA signature targeted by deregulated miRNA in long COVID

Mechanistic Class	Compounds
Signal-transduction enzyme inhibitors and modulators	5-Iodotubercidin, ZG-10 (JNK inhibitor), Cilostazol, L-744832
Vitamin D metabolite/analog	Secalciferol (24 R,25-dihydroxyvitamin D3)
G-protein coupled receptor/cholinergic-neuropeptide receptor modulation	Clidinium, LY-288513, compounds that downregulate the kisspeptin (KISS1) receptor
Antiviral nucleoside analog	Vidarabine
Lipid metabolism modulator	Torcetrapib

The most discriminative variable from our analyses was hsa-miR-484/hsa-miR-126-3p, which showed moderate diagnostic accuracy for long COVID with an AUC of 0.676 in external validation analysis. This miRNA ratio integrates two biologically meaningful signals and may reflect an imbalance between mitochondrial stress signaling and endothelial repair mechanisms: hsa-miR-484 has been reported in the context of mitochondrial and endothelial dysfunction, while hsa-miR-126-3p plays a central role in angiogenesis and vascular inflammation⁴⁹. In our dataset, a lower ratio was strongly associated with long COVID. This could suggest either downregulation of hsa-miR-484 in response to apoptotic stimulation or a relative increase in hsa-miR-126-3p expression, potentially a compensatory response to endothelial injury or dysfunction in patients with persistent symptoms after acute COVID-19 infection. This would be in line with a recent pediatric study that reported a distinct protein signature from blood plasma in children, which was characterized by ongoing inflammatory and pro-angiogenic signals⁵⁰. These discoveries suggest that the miRNA ratio may reflect underlying pathophysiological mechanisms relevant to long COVID.

At a set classification threshold of 0.8, our logistic regression model with hsa-miR-484/hsa-miR-126-3p, hsa-miR-345-5p/hsa-miR-223-3p and age and sex as covariates did not transfer well to the external validation cohort. However, despite reduction of AUC (from 0.760 to 0.676), the miRNA ratios showed consistent directionality for long COVID patients compared to controls (Fig. 4E, Fig. 4F). Performance metrics of our model remain similar to the AUC values reported by Szögi et al., ranging from 0.715 to 0.758, in their investigation of proposing the use of circulating cell-free mitochondrial DNA in cerebral spinal fluid as a new biomarker for long COVID⁵¹. Notably, the distributions for both ratios were shifted toward more negative values in the external cohort for both controls and long COVID (e.g., miR-484/miR-126-3p control median -4.14 in external vs. -2.77 in training, miR-345/miR-223 control median -9.14 vs. -8.32), consistent with a cohort effect on the measurement scale. This systematic shift provides a plausible explanation for the calibration intercept < 0 and

the very high predicted probabilities in validation when applying fixed coefficients and the training-derived cutoff. However, the preserved directionality of lower miRNA ratios in long COVID consistent across cohorts suggests these ratios may still capture a biologically relevant signal and could remain useful as relative biomarkers, provided that the model may be transferred through better methods of recalibration, batch harmonization, or cohort-specific thresholding before clinical application.

Past studies have mainly evaluated miRNAs for their prognostic value in hospitalized COVID-19 patients during the acute infection phase. In this setting, using miRNA ratios brings the advantage of technical variability stabilization compared to singular miRNA values, while integrating correlated biology. Molinero et al. found the miR-199a-5p/miR-9-5p ratio from bronchial aspirate samples the most accurate for risk stratification for critically ill COVID-19 patients after initiation of invasive mechanical ventilation to predict intensive care unit (ICU) mortality⁵². In another exploratory study of hospitalized COVID patients, more severe courses of disease, defined as ICU stay, were linked to higher miR-223-3p and miR-126-3p levels⁵³. In our long COVID cohorts, both miR-126-3p and miR-223-3p likewise emerged as recurring components of the most informative ratios (hsa-miR-484/hsa-miR-126-3p and hsa-miR-345-5p/hsa-miR-223-3p), supporting their broader relevance across the COVID-19 disease spectrum. Although long COVID is clinically distinct from acute severe COVID-19, the repeated involvement of endothelial- and myeloid-associated miRNAs is consistent with hypotheses implicating persistent immune activation and vascular dysregulation in post-acute disease. In this context, considering that persistent viral RNA or protein antigens have been hypothesized as pathomechanisms of long COVID, anti-inflammatory and antiviral properties of drug compounds may be of particular interest, for the low-grade inflammation disease state leading to ongoing symptoms⁵⁴.

To contextualize the signature biologically and generate therapeutic hypotheses, we integrated miRNA target prediction with LINCSeq L1000 perturbational signatures and constructed a miRNA-gene target-drug network with compounds that were significantly perturbing downregulation of

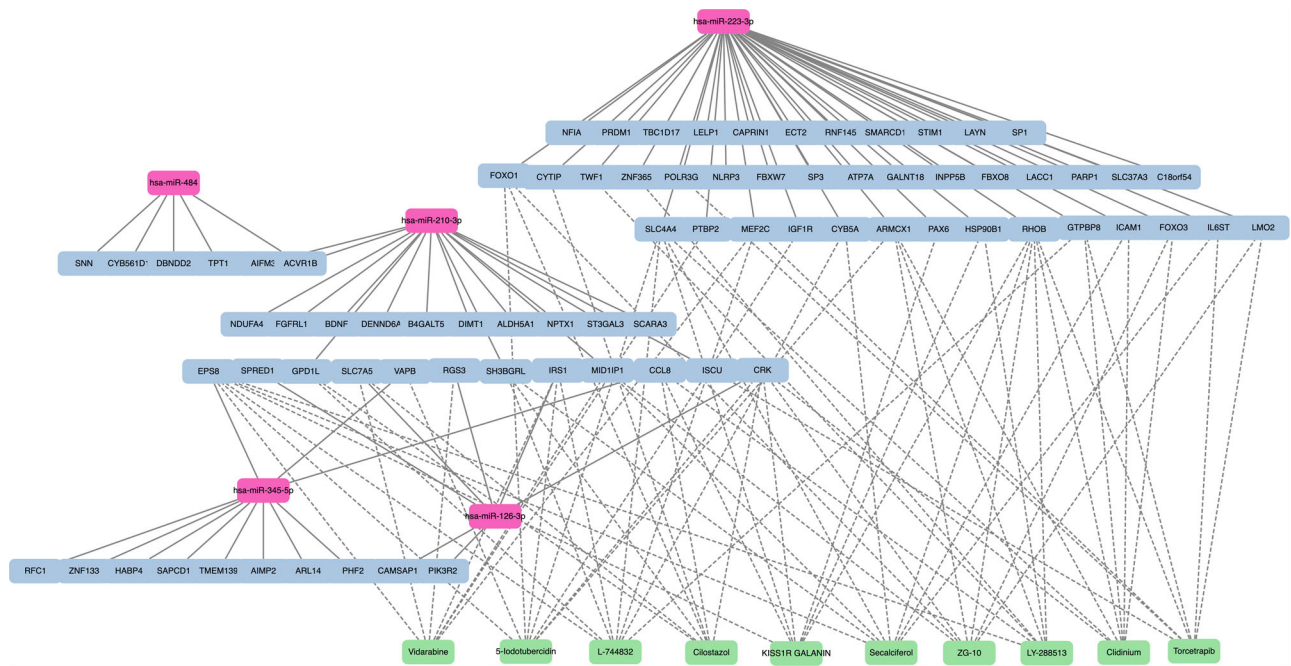


Fig. 5 | miRNA-mRNA-pathway-drug target network. Pink nodes show DE-downregulated miRNAs in LC compared to Healthy, blue nodes represent target genes of these miRNAs, and green nodes depict the drug compounds that are predicted to downregulate the gene signatures (connected by dashed lines).

the gene signature (Fig. 5). Interestingly, activin receptor type-1B (ACVR1B, also known as ALK4), was the target gene of both hsa-miR-484 and hsa-miR-210-3p. An experimental study found that increased ACVR1B levels promoted the activation of the TGF-beta signaling pathway by increasing levels of the phospho-Smad2/3, which increased the expression of the transcription factor C/EBP α and finally promoted monocyte/macrophage differentiation⁵⁵. Upregulation of TGF-beta signaling and TGF-beta receptor signaling found in pathway enrichment possesses pleiotropic effects in immunity, apoptosis and angiogenesis⁵⁶. Given prior reports of monocyte activation and tissue-resident macrophage changes in long COVID, dysregulation of miRNAs targeting ACVR1B may contribute to persistent low-grade inflammation or fibrotic signalling⁵⁷.

Exploratory drug target analysis showed some compounds were predicted to reverse the miRNA-mRNA signature, such as 5-iodotubercidin, secalciferol or ZG-10 (c-Jun N-terminal kinase (JNK) inhibitor). 5-Iodotubercidin is a selective adenosine kinase (ADK) inhibitor that has been shown to attenuate endothelial activation in inflammatory settings by suppressing TNF- α -induced adhesion molecule expression⁵⁸. In addition, 5-Iodotubercidin has been reported to inhibit SARS-CoV-2 RNA synthesis in a cell-based RNA replicase reporter system, with the antiviral effect linked to ADK activity rather than direct polymerase binding⁵⁹. Secalciferol is a vitamin D3 metabolite; vitamin D metabolites in general have been attributed with immunomodulatory and anti-inflammatory effects across immune cells, providing a plausible rationale for its appearance among connectivity hits⁶⁰. ZG-10, a c-Jun N-terminal kinase (JNK) inhibitor, can target JNK, which has a role in inflammatory signaling and was shown to support coronavirus replication cycles⁶¹. Still, most compounds highlighted by connectivity mapping remain preclinical and should only be viewed as starting points for mechanistic validation rather than candidates for immediate off-label use. In the real-world setting, a large patient-reported treatment survey with 1800 long COVID patients reported on the perceived benefit from several off-label symptomatic therapies, such as using IVIG/IgG, antihistamines, beta blockers, or pyridostigmine⁶². Reported responses varied across symptom-defined subgroups, underscoring that off-label management is best individualized rather than applied uniformly⁶².

This study has some limitations. First, the cohort was recruited from a single specialized outpatient clinic and included only adults, which may limit generalizability to broader populations, including pediatric patients

with long COVID. The control design did not include time-matched post infection controls, so some differences may reflect the post infection status rather than long COVID-specific biology. Although subgroup classification was grounded in objective clinical, imaging, and functional criteria, symptom overlap, and subclinical manifestations may have influenced group allocation. RNA-seq and qPCR analyses were restricted to blood-derived miRNA profiles, which may not fully capture tissue-specific regulatory processes, and qPCR experiments had to be performed in batches, which could have introduced batch effects. Further, the single time point design does not allow for following the longitudinal changes in the identified miRNA ratios. Longitudinal studies will therefore be necessary to characterize the temporal dynamics of these miRNA measures and clarify how the associated miRNA-mRNA-regulated pathways evolve over time. The sample size, particularly within subgroups, may constrain statistical power. Nonetheless, stringent measures were implemented throughout, such as the inclusion of spike-in controls for RNA extraction and hemolysis indices to minimize pre-analytical variability, rigorous quality control protocols for next-generation sequencing and RT-qPCR analyses, and target prediction based on validated databases with conservative thresholds to enhance specificity. While these strategies strengthen the robustness of our findings, functional validation of miRNA-target interactions and drug target predictions based on in silico enrichment analyses is still required. Thus, the findings of this study should be interpreted with due caution, while acknowledging the comprehensive and systematic approach applied across clinical, molecular and statistical domains.

Conclusion

We identified differentially expressed miRNAs in long COVID patients following initial mild COVID-19 disease, with hsa-miR-484, hsa-miR-345-5p and hsa-miR-126-3p being implicated across multiple detection modalities. Despite lower discrimination and miscalibration in external validation, both miRNA ratios from the model are directionally consistent in being lower compared to the controls of their cohort. This suggests a reproducible association with long COVID, even if systematic shifts may still be present depending on batch. Thus, hsa-miR-484/hsa-miR-126-3p and hsa-miR-345-5p/hsa-miR-223-3p may capture biologically relevant signal but are sensitive to cohort and pre-analytical differences. Further standardized diagnostic investigations and large prospective studies

mitigating sampling bias are required to confirm the diagnostic and biological relevance of the identified miRNAs.

Data availability

The miRNA counts data from the discovery phase are provided as raw reads in Suppl. Data 1. Numerical source data for the Figs. 2A, 3A, 3C, 4E, 4F of this article can be found in the Suppl. Data 2. All other relevant data to this study can be requested from the corresponding author upon reasonable request, with responses typically provided within two weeks.

This study was conducted as part of the POSTCOV registry, which is registered on ClinicalTrials.gov (NCT05398952). The full study protocol has not been published, but information can be found on the official funding website of the Austrian Science Fund (<https://www.fwf.ac.at/en/research-radar/10.55776/KLI1064>).

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Author contributions

M.G. conceived and supervised the study, provided resources, funding acquisition and contributed to data interpretation. EmH developed the methodology, performed the experiments, analysed and interpreted the data, created the visualizations, and wrote the original draft. D.L. and K.S. contributed to sample processing and RT-qPCR experiments and supported data curation. En.H., K.M.-Z. and N.K. contributed to participant recruitment, sample handling, and laboratory investigations. A.S. provided methodological support and resources. M.R. contributed to participant recruitment and coordination. C.N. and T.A.Z. provided resources and supervision. KH oversaw project administration and coordinated the work. All authors contributed to review and editing.

Competing interests

The authors declare no competing interests.

Additional information

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